Deep Learning and Causal AI for Estimation

Lecture 4: Downstream Regression

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From AI to Econometrics

We have seen various strategies to generate economic variables with AI systems.

The end goal for most empirical economists is to use these generated variables in downstream regression analysis.

Important lessons:

1. Measurement error is a feature of AI-generated variables
2. Zero-shot/few-shot learning with generic models can underperform older approaches
3. Recent model development focuses on verifiable rewards

In light of these issues, how can one perform valid statistical inference?

Issue is generic; applies to causal and non-causal models.

Generic Setup

Consider the regression model

$$Y_i = \boldsymbol{\gamma}^T \boldsymbol{\theta}_i + \boldsymbol{\alpha}^T \mathbf{q}_i + \varepsilon_i, \qquad \mathbb{E}\left[\varepsilon_i \middle| \boldsymbol{\theta}_i, \mathbf{q}_i\right] = 0,$$

where Y_i is a potentially missing outcome variable, $\boldsymbol{\theta}_i$ are potentially missing covariates, and $\mathbf{q}_i$ are observed controls.

Unstructured dataset $\mathbf{x}_i$ is availble to impute missing data.

Examples:

Y_i is a latent belief, $\theta_i \in \{0, 1\}$ is an observed information treatment, and $\mathbf{x}_i$ is a written survey response.

Y_i is an observed macro outcomes at time i, θ_i is latent sentiment, and $\mathbf{x}_i$ is a collection of newspaper articles.

Naive Inference

The most common strategy is a two-step approach:

1. Use an ML/AI classifier with $\mathbf{x}_i$ as input to generate predicted $\hat{Y}_i$ or $\hat{\boldsymbol{\theta}}_i$, e.g. by using the methods discussed in previous slides
2. Plug predicted variables into downstream regression, compute point and interval estimates as if they were observed

Two-step approach yields invalid inference.

Measurement error is non-classical and cannot be corrected with standard formulas.

How should one proceed?

Outline

Solutions with Validation Data

Solutions without Validation Data

Definition of Validation Data

Growing literature relies on a validation sample: a subset of observations where the truth is observed alongside the AI-generated prediction $\rightarrow$ truth and prediction observed jointly on the same units.

Missing regressor setting ($\boldsymbol{\theta}_i$ generated):

Validation sample: $(Y_i, \hat{\boldsymbol{\theta}}_i, \boldsymbol{\theta}_i, \mathbf{q}_i)_{i=1}^{m}$

Main sample: $(Y_i, \hat{\boldsymbol{\theta}}_i, \mathbf{q}_i)_{i=m+1}^{n+m}$

Missing outcome setting (Y_i generated):

Validation sample: $(Y_i, \hat{Y}_i, \boldsymbol{\theta}_i, \mathbf{q}_i)_{i=1}^{m}$

Main sample: $(\hat{Y}_i, \boldsymbol{\theta}_i, \mathbf{q}_i)_{i=m+1}^{n+m}$

Valid inference can be performed using just the validation sample: AI useful to improve efficiency, not correct bias.

Prediction-Powered Inference

[Angelopoulos et al., 2023] introduce Prediction-Powered Inference (PPI) framework for inference with missing outcome variables.

Compute an estimate of a quantity of interest using the main sample.

Use the validation dataset to construct a ‘rectifier’ that adjusts the initial estimate.

Does not require specifying a particular form of measurement error in the classifier.

See [Ludwig et al., 2025] for a related discussion.

Example for Estimating Mean

$$\hat{\mu}^{PP} = \underbrace{\frac{1}{n} \sum_{i=m+1}^{n+m} \hat{Y}_i}_{\text{imputation (main)}} - \underbrace{\frac{1}{m} \sum_{i=1}^{m} \left(\hat{Y}_i - Y_i \right)}_{\text{rectifier (validation)}}$$

Unbiasedness requires that $\mathbb{E}\hat{Y}_i$ is the same on main and validation samples.

$\frac{n}{m} \to \rho$ so that the influence of the rectifier doesn't disappear asymptotically.

Efficiency gain relative to using validation sample only if

$$\frac{\sigma^2_{\hat{Y}-Y}}{m} + \frac{\sigma^2_{\hat{Y}}}{n} < \frac{\sigma^2_{Y}}{m}$$

Similar comments apply to regression coefficient estimates, see also [Egami et al., 2023].

Missing Regressors

Consider now the missing regressor setting and a simplified model where the regression model is

$$Y_i = \gamma\theta_i + \varepsilon_i$$

and suppose the researcher runs OLS of Y_i on $\hat{\theta}_i$.

$$\hat{\gamma}^{\mathrm{OLS}} \xrightarrow{p} \gamma \frac{\mathsf{Cov}(\hat{\theta}_i, \theta_i)}{\mathsf{V}(\hat{\theta}_i)}$$

The distortion term is the coefficient from regressing θ_i on $\hat{\theta}_i$.

Two-Stage Least Squares

The bias formula suggests a 2SLS strategy [Fong and Tyler, 2021, for example]:

1. Regress θ_i on $\hat{\theta}_i$ in the validation sample, obtain coefficient $\hat{\beta}$.
2. Regress Y_i on $\hat{\beta}\hat{\theta}_i$ in the main sample.

$$\hat{\gamma}^{\mathrm{2SLS}} \xrightarrow{p} \gamma \cdot \frac{\mathsf{Cov}(\beta\hat{\theta}_i, \theta_i)}{\mathsf{V}(\beta\hat{\theta}_i)} = \gamma \cdot \frac{1}{\beta} \cdot \frac{\mathsf{Cov}(\hat{\theta}_i, \theta_i)}{\mathsf{V}(\hat{\theta}_i)}$$

Where $\beta = \mathsf{Cov}(\hat{\theta}_i, \theta_i)/\,\mathsf{V}(\hat{\theta}_i)$. This implies:

$$\hat{\gamma}^{\mathrm{2SLS}} \xrightarrow{p} \gamma.$$

Resulting estimate is consistent if joint distribution of $Y_i, \theta_i, \hat{\theta}_i$ is stable across validation and main samples.

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Validation Data in Economics

Validation dataset typically requires human labels.

But use of ML/AI typically motivated:

1. Large n
2. Large cost of human labels

Unclear how a validation dataset of appropriate size can be constructed in this environment.

Deeper question: unclear that a "true" $\boldsymbol{\theta}_i$ can be observed.

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Need for valid inference without (much) validation data!

[Battaglia et al., 2025] assumes the measurement error is of the same asymptotic order as sampling uncertainty in the regression model.

Baseline Case: AI Indices

Several influential works generate indices by classifying documents + aggregating: [Baker et al., 2016], [Caldara and Iacoviello, 2022], [Gorodnichenko et al., 2023].

The researcher observes a vector X_i containing the number of units classified in each of the J categories.

Natural data-generating process is

$$X_i | (C_i, \theta_i) \sim \text{Multinomial}(C_i, B\theta_i), \tag{1}$$

where

- C_i is number of classified units.
- θ_i is a J-dimensional probability vector from which true label is drawn
- b_{ij} denotes the probability that a unit with true label j is assigned label i.

Naive estimator of θ_i is $P_i = X_i / C_i$, i.e. empirical share. True θ_i *never* observed!

Measurement error arises from misclassification error (B) and sampling error (C_i).

Example

In [Gorodnichenko et al., 2023], C_i is the number of FOMC statement paragraphs at meeting i.

$J = 3$: dovish/neutral/hawkish

Hawkish index is (normalized) net count of hawkish paragraphs: $X_{i,H} - X_{i,D}$.

Fit regression model of target factor [Gürkaynak et al., 2005] on hawkish index and shadow short rate [Wu and Xia, 2016].

Simulation Calibrated to [Gorodnichenko et al., 2023]

	$\delta = 0$			$\delta = 1$			$\delta = 2$		
n	$c = 1$	$c = 2$	$c = 4$	$c = 1$	$c = 2$	$c = 4$	$c = 1$	$c = 2$	$c = 4$
	Bias of OLS Estimator of γ								
200	-0.207	-0.034	0.082	-0.177	0.023	0.173	-0.179	0.065	0.236
400	-0.201	-0.029	0.078	-0.193	0.017	0.163	-0.178	0.066	0.237
800	-0.204	-0.034	0.074	-0.185	0.014	0.159	-0.181	0.06	0.242
1600	-0.206	-0.035	0.082	-0.186	0.014	0.159	-0.177	0.063	0.241
3200	-0.206	-0.033	0.081	-0.191	0.014	0.161	-0.181	0.058	0.24
	Coverage of 95% Confidence Intervals for γ								
200	0.714	0.937	0.922	0.786	0.931	0.839	0.805	0.941	0.78
400	0.514	0.933	0.902	0.583	0.943	0.753	0.677	0.92	0.622
800	0.203	0.92	0.849	0.359	0.955	0.588	0.434	0.894	0.333
1600	0.028	0.888	0.7	0.08	0.95	0.333	0.17	0.845	0.083
3200	0.001	0.864	0.505	0.004	0.93	0.072	0.011	0.774	0.002

Bias Correction

Step One: Purge Classification Error

From the multinomial DGP, $\mathbb{E}[P \mid \Theta, C] = B\Theta$.

Estimate of true label probabilities is $\hat{\Theta} = \hat{B}^{-1}P$.

Where to obtain $\hat{B}$? (1) confusion matrix; (2) vertex hunting

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Step Two: Bias Correction of Downstream Regression

Fit OLS with $\hat{\theta}_i$ as regressor.

Assumption: $\sqrt{n} \times \mathbb{E}[C_i^{-1}]$ converges to a constant $\kappa \in [0, \infty)$

Distribution of OLS coefficients has first order bias proportional to κ but standard errors are correct.

Use analytic expression for bias to correction (formula in [Battaglia et al., 2025].)

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Bias correction implemented in `ValidMLInference` [Christensen and Hansen, 2026].

Results

Table: Effect of Fed Sentiment on Policy Path

	Two-Step (Raw Freq.)	Two-Step (Estimate)	Bias-Corrected
Hawkish	0.080	0.063	0.090
	[−0.006, 0.165]	[−0.012, 0.138]	[0.015, 0.165]
Dovish	−0.048	−0.046	−0.098
	[−0.159, 0.063]	[−0.145, 0.053]	[−0.197, 0.001]
Shadow Rate	−0.005	−0.005	−0.009
	[−0.014, 0.004]	[−0.013, 0.004]	[−0.018, −0.001]

95% confidence intervals in brackets. Outcome is the GSS path factor measuring expected future policy rates. Sample covers 199 FOMC meetings from 1995–2023. The first column measures sentiment with raw label frequencies. The second uses estimates of hawkish and dovish labels probabilities. The third applies additive bias correction to the second.

Additional Case: Binary Label Imputation

Suppose instead that θ_i is binary.

Measurement error assumption is $\lim_{n\to\infty} \sqrt{n} \times \text{FPR} = \kappa \in [0, \infty)$

Bias correction requires estimation of FPR.

Can either come from known performance on external dataset, or by manual inspection of m observations where $\frac{n}{m^2} \to 0$.

Hansen Lambert Bloom Davis Sadun Taska (WP, 2023)

- Consider $n = 16,315$ SD food+accom sector (NAICS code 72) job postings from January 2022
- Regress log wages Y_i on ML-generated remote work indicator $\hat{\theta}_i$
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- Fixed effects for SOC code (job type) and full/part time
- For bias correction, use estimate $\widehat{FPR} \approx 0.009$.
- For joint estimation, use three-component Gaussian mixture for errors $\varepsilon_i | \theta_i$

Bias Correction with Minimal Human Effort

Advantage 1: Smaller Auxiliary Dataset

PPI: bias correction when m and n are comparable.

We estimate FPR with $m = 1000$. $n/m = 16$, $n/m^2 = 0.016$.

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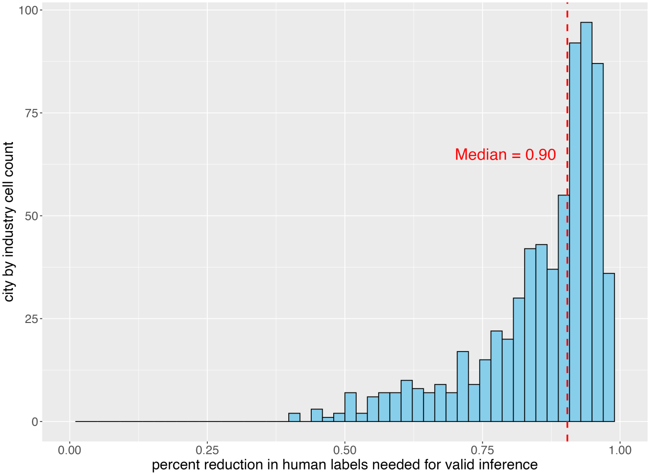
Advantage 2: Only Need Partial Labeling

PPI: build full validation dataset by inspecting each posting.

This paper: only examine labeled "ones", 26 in this dataset.

Distribution of Reduction in Human Labels



Results

Table: Effect of Remote Work on Log Salary

	Two-Step	Bias-Corrected
Remote Work Indicator	0.364	0.641
	[0.322, 0.406]	[0.446, 0.836]
Occupation FE	Yes	Yes
Employment Type FE	Yes	Yes

95% confidence intervals in brackets. The bias-corrected column uses `ols_bcm` with FPR $= 0.009$ estimated from an external sample of $m = 1{,}000$ hand-labeled job postings.

References I

Angelopoulos, A. N., Bates, S., Fannjiang, C., Jordan, M. I., and Zrnic, T. (2023).
Prediction-powered inference.
Science, 382(6671):669–674.

Baker, S. R., Bloom, N., and Davis, S. J. (2016).
Measuring Economic Policy Uncertainty.
The Quarterly Journal of Economics, 131(4):1593–1636.

Battaglia, L., Christensen, T., Hansen, S., and Sacher, S. (2025).
Inference for Regression with Variables Generated by AI or Machine Learning.

Caldara, D. and Iacoviello, M. (2022).
Measuring Geopolitical Risk.
American Economic Review, 112(4):1194–1225.

Christensen, T. and Hansen, S. (2026).
Performing Valid Inference with AI/ML-Generated Covariates: A Guide for Empirical Practice.
AEA Papers and Proceedings, 116:92–97.

References II

Egami, N., Hinck, M., Stewart, B. M., and Wei, H. (2023).
Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models.
In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, NIPS '23, pages 68589–68601, Red Hook, NY, USA. Curran Associates Inc.

Fong, C. and Tyler, M. (2021).
Machine Learning Predictions as Regression Covariates.
Political Analysis, 29(4):467–484.

Gorodnichenko, Y., Pham, T., and Talavera, O. (2023).
The Voice of Monetary Policy.
American Economic Review, 113(2):548–584.

Gürkaynak, R. S., Sack, B., and Swanson, E. (2005).
Do actions speak louder than words? The response of asset prices to monetary policy actions and statements.
International Journal of Central Banking, 1:55–93.

Ludwig, J., Mullainathan, S., and Rambachan, A. (2025).
Large Language Models: An Applied Econometric Framework.

References III

Wu, J. C. and Xia, F. D. (2016).
Measuring the macroeconomic impact of monetary policy at the zero lower bound.
Journal of Money, Credit and Banking, 48(2-3):253–291.